

Computing Fundamental

Experiment # 09

If..else

1. Objective/s:

- If Statement
- If-else Statement
- Nested if Statement
- Nested if else Statement
- Relational Operators
- Logical Operators

2. Conditional Statement

The conditional statements are used to execute a set of statements after testing a condition. The conditional statements are also called selection statements

2.1.The “if” Statement:

The “**if statement**” evaluates a condition. If the given condition is **true**, the statement (or set of statements) following the “**if statement**” is executed. But if the given condition is **false**, the statement (or set of statements) following the “if statement” is ignored and the control transfers to the next statement.

2.1.1. Syntax of the “if Statement” is:

```
If (condition)
Statement – 1;
Statement – 2;
```

To execute a set of statements following the “**if Statement**”, the set of statements is enclosed in curly braces.

```
If (condition)
{
Statement – 1;
Statement – 2;
Statement – m
}
Statement – n
```

2.1.2. Example Code

```
int main()
{
int a;
Printf("\n Enter the number = ");
Scanf("%d",&a);
if (a%3==0)
{
Printf("\n The number is divisible by 3");
}
Printf("\n The number is not divisible by 3");
}
```

2.2.The “if..else” Statement

In this statement, one condition and two blocks of statements are given. Either one of the two blocks of statements is executed after evaluating a condition. If the given condition is **true**, the first block of statements is executed. But, If the given condition is **false**, the first block of statements is ignored and the second block following the else is executed.

2.2.1. Syntax of the “if else Statement”

```
    If (condition)
        Statement – 1;
    else
        Statement – 2;
```

In the case of multiple statements, the syntax is written as:

```
    if (condition)
        Statement – 1;
        Statement – 2;
        Statement – m;
    }
    else
    {
        Statement – 1;
        Statement – 2;
        Statement – n
    }
```

2.2.2. Example Code

```
void main()
{
int a;
Printf("\n Enter the number = ");
Scanf("%d",&a);
```

```
if (a>100)
Printf("\n The number is greater than 100";
else
Printf("\n The number is less than 100";
}
```

2.3.Nested If Statement

When an “if statement” is used within another “if statement”, it is called the “nested if statement”. The “nested if statement” is used for multi way decision-making. The syntax of “nested if statement” is:

```
if (condition-1)
{
if (condition-2)
{
Statement-2;
}
Statement-3;
}
```

2.3.1. Example Code

```
void main()
{
int a,b,c;
Printf("Enter first Number = ");
Scanf("%d", &a);
Printf("\n Enter second Number = ");
Scanf("%d", &b);
Printf("\n Enter third Number = ");
Scanf("%d", &c);
if(a>b){
    if(a>c){
        Printf("\n\n 1st Number is greater ");    }
    else
        Printf("\n\n Third number is greater ");    }
else
    if (b>c){
        Printf("\n\n Second number is greater ");    }
    else
        Printf("\n\n Third number is greater ");    }
```

3. Logical Operators

The logical operators are used to combine relational expressions or relational conditions.

The output of a logical expression is also in a logical form. Its value is either true or false.

In C++, following logical operators are involved:

- && → (AND Operator)
- || → (OR Operator)
- ! → (NOT Operator)

3.1.Operators & If else Statement

Conditions can be combined by operators && (AND), || (OR) or negated by !(NOT).

3.2.Example Code:

For example, we might want to know if the value of variable x is between 10 and 20. This is actually two conditions: we need to know if x is greater than 10, and also whether x is less than 20.

```
void main()
{
int n;
Printf("Enter a number: ");
Scanf("%d", &n);
if (n >10 && n< 20)
    printf("Your value is between 10 and 20");
else
    printf ("You value is not between 10 and 20");
}
```

4. Tasks

5. Write a program to input a number from the key board and then print the message “It is an odd number” if the given number is odd number. Else print “It is an even number”. (A number is even it is divisible by 2)
6. Write a program to input two numbers and then find out whether these numbers are equal or different
7. Write a program in C++ to input marks obtained by a student in a subject. The total marks are 100. Find out the grade of the student by using the if-else nested structure. The grade is:
 - If marks are equal to or greater than 90, grade is A.
 - If marks are equal to or greater than 70 and less than 90, grade is B.
 - If marks are equal to or greater than 50 and less than 70, grade is C.
 - If marks are less than 50, grade is “F”.